

Standard Model of Elementary Particles

three generations of matter
(fermions)

interactions / force carriers
(bosons)

I

II

III

mass
charge
spin

QUARKS

LEPTONS

$\approx 2.16 \text{ MeV}/c^2$

$\frac{2}{3}$
 $\frac{1}{2}$
u

up

$\approx 1.273 \text{ GeV}/c^2$

$\frac{2}{3}$
 $\frac{1}{2}$
c

charm

$\approx 172.57 \text{ GeV}/c^2$

$\frac{2}{3}$
 $\frac{1}{2}$
t

top

$\approx 4.7 \text{ MeV}/c^2$

$-\frac{1}{3}$
 $\frac{1}{2}$
d

down

$\approx 93.5 \text{ MeV}/c^2$

$-\frac{1}{3}$
 $\frac{1}{2}$
s

strange

$\approx 4.183 \text{ GeV}/c^2$

$-\frac{1}{3}$
 $\frac{1}{2}$
b

bottom

$\approx 0.511 \text{ MeV}/c^2$

-1
 $\frac{1}{2}$
e

electron

$\approx 105.66 \text{ MeV}/c^2$

-1
 $\frac{1}{2}$
 μ

muon

$\approx 1.77693 \text{ GeV}/c^2$

-1
 $\frac{1}{2}$
 τ

tau

$< 0.8 \text{ eV}/c^2$

0
 $\frac{1}{2}$
 ν_e

electron
neutrino

$< 0.17 \text{ MeV}/c^2$

0
 $\frac{1}{2}$
 ν_μ

muon
neutrino

$< 18.2 \text{ MeV}/c^2$

0
 $\frac{1}{2}$
 ν_τ

tau
neutrino

0

0

1

g

gluon

$\approx 125.2 \text{ GeV}/c^2$

0

0

H

higgs

0

0

1

γ

photon

$\approx 91.188 \text{ GeV}/c^2$

0

1

Z

Z boson

$\approx 80.3692 \text{ GeV}/c^2$

± 1

1

W

W boson

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS